IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID: 25-26J-331

1.Topic (12 words max)

Real-Time AI-IoT Elephant Detection and Acoustic Deterrent for Sri Lankan Railways

2.Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3.Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5.Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Elephant-train collisions have become a significant and tragic problem in Sri Lanka, especially on railway lines that intersect traditional elephant migration routes. These collisions frequently result in the death or serious injury of elephants, substantial damage to trains, and endanger the lives of passengers and train staff [1]. According to the Department of Wildlife Conservation (DWC), dozens of elephants die annually in train accidents across the country, with major hotspots identified in areas such as Habarana, Galgamuwa, and Minneriya [2].

The primary causes of these collisions include the absence of real-time detection systems, poor night-time visibility, and the lack of effective early warning mechanisms for train drivers [3]. Existing mitigation efforts, such as fencing along railway tracks and human patrolling, have shown limited effectiveness due to coverage gaps and slow response times [4]. Consequently, there is an urgent need for an intelligent, automated system that can monitor known elephant crossing points, detect elephants reliably, assess the risk of collisions in real-time, and trigger protective measures promptly.

The aim of this research is to develop a novel system integrating Artificial Intelligence (AI) and the Internet of Things (IoT) technologies to address the problem. This system is designed to continuously monitor elephant crossing locations through a network of cameras and sensors, applying computer vision algorithms to detect and classify elephants. The system distinguishes elephants from other animals in the environment and classifies them into categories such as individual elephants, herds, or family units including calves, by accurately counting the number of detected animals [5].

Additionally, the system analyzes elephant body postures to assess behavioral states, categorizing them as normal, frightened, or aggressive. This behavioral classification is crucial for deciding the appropriate intervention method [6]. For instance, if elephants are aggressive or belong to a vulnerable family group (such as a three-part herd with adult-calf-adult composition), certain deterrents such as bee sound circuits are withheld to prevent escalation of stress or harm to the animals [7].

Once elephants are detected and classified, a protective mechanism is triggered, which emits bee sounds designed to scare elephants away from the railway tracks. The activation of this mechanism is conditional based on the elephants' behavior and proximity to human settlements, ensuring ethical and non-invasive intervention [8]. Simultaneously, the system uses GPS and real-time tracking data to calculate the distance between elephants and approaching trains, continuously assessing the risk level.

Risk management is dynamically conducted with the system sending alerts to train drivers. Depending on the risk assessment, alerts vary from pre-warning notifications to emergency warnings when a collision is imminent [9]. To facilitate real-time communication and rapid decision-making, a mobile application interface is being developed, providing train operators with immediate and clear warnings [10].

This AI-IoT based solution presents a scalable, data-driven, and adaptive method to mitigate elephant-train collisions in Sri Lanka. Unlike previous efforts, it combines automated detection, behavioral analysis, ethical deterrence, and real-time risk communication, all tailored to the unique ecological and infrastructural conditions of Sri Lankan railway lines. This approach not only helps save elephant lives and reduce train damages but also contributes to mitigating the broader human-elephant conflict in the region through a non-invasive and responsible use of technology [11].

References

1. Department of Wildlife Conservation, Sri Lanka – Annual Elephant Mortality Report, 2023.
2. Wijesinghe, A., et al., "Elephant Hotspots and Collision Statistics in Sri Lanka," Journal of Wildlife Conservation, 2022.
3. Perera, S., "Challenges in Night-Time Elephant Detection Along Railways," Sri Lanka Railway Safety Review, 2021.
4. Fernando, P., "Efficacy of Fencing and Patrols for Elephant-Tracks," International Journal of Wildlife Management, 2020.
5. Silva, D., & Kumar, R., "AI-Based Elephant Detection Using Computer Vision," Proceedings of the Sri Lanka Tech Symposium, 2024.
6. Rajapaksha, M., "Elephant Behavior Classification Using Posture Analysis," Asian Journal of Animal Behavior, 2023.
7. De Silva, K., "Non-Invasive Deterrents for Elephant Conflict Mitigation," Wildlife Ethical Journal, 2022.
8. Gunawardana, T., "Bee Sound Deterrents: Effectiveness and Risks," Sri Lanka Environmental Review, 2023.
9. Jayawardene, H., "Real-Time Risk Management Systems for Railway Safety," Transportation Safety Journal, 2024.
10. Fernando, L., "Mobile Alert Applications for Train Operators in Wildlife Zones," ICT for Conservation, 2023.
11. Perera, N., & Silva, M., "AI and IoT in Human-Elephant Conflict Mitigation: Case Study Sri Lanka," Environmental Technology Review, 2024.

6.Brief description of the nature of the solution including a conceptual diagram (250 words max)

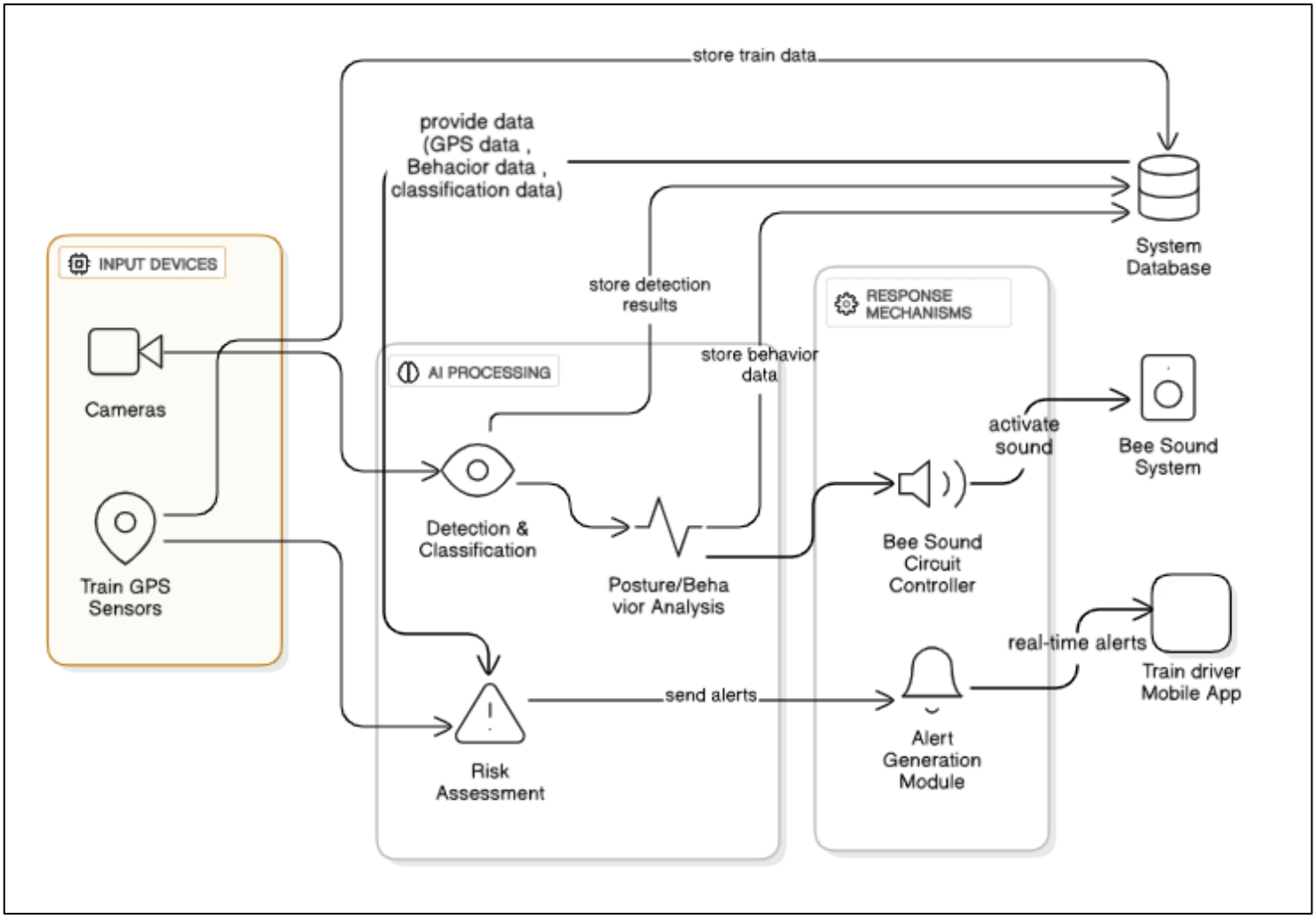
This AI- and IoT-powered system is designed to prevent elephant-train collisions in Sri Lanka by detecting elephants near railway crossings, assessing the risk, and initiating appropriate response actions. Cameras installed at elephant-train crossings continuously monitor the presence of elephants. Using the camera view, the system distinguishes elephants from other animals and, after counting the number of elephants, categorizes them as individual elephants, herds, or three-part herds (adult-calf-adult).

The system also uses posture analysis to identify elephant behavior - normal, frightened, or aggressive. Based on behavior, classification, and proximity to human settlements, the bee colony selectively activates a noise suppression system. The system mimics the sound system of a natural bee colony to repel elephants. However, the sound is not used when elephants are showing aggressive behavior, are part of a three-part herd, or are close to a village to avoid negative consequences.

At the same time, the system calculates the distance to the train using GPS data and also assesses the risk. The assessed risk is communicated through an alert, and if elephants are encountered within 1 km of an approaching train, an emergency warning is sent to the driver, as there may not be enough time to take preventive measures. For distances greater than 1 km, the warnings inform the driver to slow down and prepare, and real-time monitoring continues to confirm whether the elephants are still present. At 1 km, an emergency warning is sent to the driver if the elephants are still on the track, and if not, the driver is informed that the elephants have left. A mobile application is being developed for this.

This integrated approach balances conservation, railway safety, and community well-being. (Conceptual diagram -Figure 1.0)

Figure 1.0


store train data
provide data
(GPS data ,
Behacior data ,
classification data)
System
Database
INPUT DEVICES
Cameras
Train GPS
Sensors
store detection
results
AI PROCESSING
store behavior
data
RESPONSE
MECHANISMS
activate
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Bee Sound
System
Detection &
Classification
Posture/Beha
vior Analysis
Bee Sound
Circuit
Controller
real-time alerts
Train driver
Mobile App
send alerts
Risk
Assessment
Alert
Generation
Module

7.Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

The development of this AI- and IoT-based system requires multidisciplinary expertise, combining wildlife behavior analysis, artificial intelligence, computer vision, embedded systems, and railway safety protocols. Domain knowledge in elephant ethology is critical to accurately interpret behavioral states such as frightened, aggressive, or normal through posture analysis. This requires collaboration with wildlife biologists and conservationists to define posture-based behavioral cues and thresholds for safe intervention (e.g., when to activate bee sounds).

Computer vision and deep learning experts are needed to develop and train models capable of real-time elephant detection, classification (individual, herd, or three-part herd), and behavior recognition from camera feeds. Data scientists must ensure high accuracy through large, annotated datasets of elephant imagery, including diverse angles, lighting conditions, and environmental contexts.

IoT and embedded systems engineers contribute to the integration of edge computing devices (e.g., Raspberry Pi, NVIDIA Jetson) with camera modules, GPS sensors, and bee sound emitters. GPS and telemetry specialists are required to synchronize real-time train tracking and calculate dynamic distances between trains and elephants.

The system also requires geospatial and environmental data—such as elephant migration corridors, village boundaries, and train schedules—for contextual decision-making. Mobile application developers are engaged to create an interface for alert delivery and system monitoring.

High-quality datasets are vital, including:

- Annotated images/videos of elephants exhibiting different behaviors
- GPS-based train and elephant location data
- Environmental context data (proximity to villages, terrain)
- Historical elephant-train collision records

Ethical data collection methods and collaboration with local authorities (e.g., Department of Wildlife Conservation, Railway Department) are essential to ensure responsible deployment. This project leverages local ecological knowledge alongside modern AI technologies to build a robust, responsive solution.

8.Objectives and Novelty

Main Objective The main objective of this research is to develop an AI- and IoT-based system to prevent elephant-train collisions in Sri Lanka. The system detects elephants near railway tracks, analyzes their behavior, predicts the risk of collision using GPS data, and activates protective measures like sound deterrents and driver alerts. This helps protect both elephants and humans in a safe and ethical way.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22563682 PATIPOLAARACHCHI P.A.T.M.	**Context-Aware Protection and Alert Function.** - Design and implement a bee sound circuit that simulates natural bee colony sounds to safely deter elephants. - Define safety rules for activating the bee sound (e.g., do not activate near aggressive elephants, three-part herds, or human villages). - Develop an alert system to communicate risk levels and warnings to train drivers via mobile or airborne devices. - Implement layered alerts:	- Design and prototype a bee sound circuit hardware that simulates natural bee colony sounds. - Develop control software to trigger the bee sound circuit based on behavioral and detection inputs. - Define and implement rules for when the bee sound should be activated or suppressed. - Build a warning system for train drivers using SMS, mobile app notifications.	The use of the Bee Sound Deterrent is context-aware – it only activates when elephants are detected to be non-aggressive, not in three-part herds, and not near human settlements. This prevents elephants from being disturbed or provoked, reduces risks to surrounding communities, and provides trained drivers with tiered alerts (early or urgent). This proactive, data-driven warning ensures timely decisions, unlike traditional systems that rely on static or delayed responses.

	- Early warnings at safe distances. - Emergency alerts when elephants are within 1 km of the train. - Ensure timely and reliable activation of safety and warning mechanisms.	- Create different levels of alerts (early warning, emergency warning, risk management) based on distance and risk.	
IT22579522 GUNAPALA R.A.P.J.	**Behavioral State Classification Function.** - Analyze elephant body postures using posture estimation and posture recognition techniques. - Classify elephant behavior into three states: normal, fearful, and aggressive. - Establish behavioral thresholds and criteria for reliable classification. - Integrate behavioral classification with recognition results for integrated situational awareness. - Prevent the triggering of avoidance mechanisms when behaviors indicate a potential risk of escalation (e.g., aggressive or vulnerable herds).	- Research and select suitable pose recognition or pose estimation frameworks(e.g.OpenPose, MediaPipe). - Annotate datasets with elephant body postures related to behavioral states (normal, fearful, aggressive). - Develop a behavior classification model based on postures and movement patterns. - Validate model accuracy and fine-tune thresholds for behavior recognition. - Integrate behavior analysis with recognition outputs for integrated insights. .	The system uniquely applies computer vision to analyze elephant body posture, categorizing their behavior as normal, frightened, or aggressive. This behavioral insight informs safer and more humane response decisions, avoiding escalation in sensitive scenarios.

IT22289452 KUMARASENA S.T.	**Real-Time Elephant Detection and Group Classification Function.** - Develop and train an AI model to accurately identify elephants from video streams at railway crossings. - Distinguish elephants from other animals and background objects using computer vision techniques. - Count the number of identified calves, adults and elephants, distinguishing between calves and adults. - Classify identified elephants into groups: individual, herd or three-part herd (adult-calf-adult). - Ensure real-time processing capability for continuous monitoring and quick identification.	- Collect and label image/video data of elephants near railway tracks, including calves and adults. - Train and validate a deep learning model (e.g. YOLO, Fast R-CNN) to identify elephants in video frames. - Implement animal differentiation logic to filter out non-elephant identifications. - Develop algorithms to count elephants and classify group structure: individual, herd, or herd of three. - Integrate the identification module with live camera feeds for real-time monitoring. - Optimize processing speed and accuracy for deployment on edge devices (e.g. Raspberry Pi, Jetson Nano).	Unlike standard systems that simply identify the presence of animals, this method distinguishes elephants from other animals and, after counting them (elephants, calves, elephants), further categorizes them as individuals, as a herd, and as a vulnerable three-part herd (adult-calf-adult) – enabling more context-sensitive interventions.
IT22142450 YASHODA W.C.M.N.	**Risk Forecasting and Collision Prediction Function.**	- Set up GPS data acquisition for trains and detection points. - Develop algorithms to calculate real-time distances between elephants and approaching trains.	By combining real-time GPS data and behavioral analysis, the system calculates collision risk levels and creates the data needed to provide

	- Collect and process GPS data from trains and elephant detection points in real time. Train GPS Tracker. - Calculate dynamic distances between detected elephants and approaching trains. - Develop risk assessment models integrating real-time data and historical collision patterns. - Determine appropriate alert levels based on collision risk predictions. - Provide continuous updates to the alert system for accurate and timely communication to train operators.	- Collect and analyze historical collision data to model risk factors. - Build a risk assessment engine integrating GPS data and behavioral/detection outputs. - Design logic to determine alert levels based on risk prediction. - Implement continuous monitoring and update mechanisms to refresh the risk situation in real time. - Integrate forecast outputs with warning and protection systems.	trained drivers with tiered alerts (early or urgent) based on the distance between the train and the elephant.

9.Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
IT22563682 PATIPOLAARACHCHI P.A.T.M.	This task deals with implementing safety measures once elephants are detected and classified. An embedded system (e.g., microcontroller) controls a bee noise suppressor circuit, which is selectively activated based on safety conditions (non-aggressive behavior, no unit family, not near villages). It sends alerts to train drivers via wireless IoT communication when elephants are nearby. - Specialization Compatibility: This is a direct application of IoT and embedded systems. It includes hardware-software interface, GPIO control, communication protocols (e.g., GSM, LoRa), and circuit integration - core elements of the Embedded/IoT specialization.
IT22579522 GUNAPALA R.A.P.J.	This function analyzes the movements of elephants using image and video frames to assess their behavioral state. Using pose estimation frames or custom convolutional neural networks, elephants are classified as normal, fearful, or aggressive. This behavioral classification helps the system determine whether or not to trigger deterrents such as noise. - Specialization Compatibility: This component includes advanced computer vision techniques and behavioral modeling, which are key areas of AI and behavioral analytics. It requires a deep understanding of image-based behavior prediction, which is a very specialized AI task.
IT22289452 KUMARASENA S.T.	This component focuses on identifying elephants near railway tracks using AI-based object recognition techniques. A trained deep learning model (e.g. YOLOv5 or SSD) processes live camera footage to identify and distinguish elephants from other animals. The model then counts the number of elephants and classifies them as individuals, herds, or structured three-part herds (adult-calf-adult). - Specialization Compatibility: The task requires expertise in artificial intelligence, specifically computer vision. This task involves preparing datasets, training and validating models, and preprocessing images - all of which fall under the AI specialization.

IT22142450 YASHODA W.C.M.N.	This function predicts the risk of elephant-train collisions using GPS and historical data. It calculates the distance between detected elephants and approaching trains, determines the probability of collision, and creates timely alerts (early or urgent). - Specialization Compatibility: This component includes GPS data processing, risk algorithm design - aligned with data science. It requires skills in back-end logic, real-time data manipulation, and machine learning for risk prediction.

10.Supervisor details

	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.